DEPARTMENT OF MECHANICAL ENGINEERING

DESIGN OF MACHINE ELEMENTS (ME 3104)

Design & Analysis of Rigid Flanged Coupling

(Protected Type — Static Design)

Interactive CAD Workstation with Web, Python & C++ Implementations

SUBMITTED BY

Ayush Raj	240103029
Aanya Verma	240103003
Gauransh Diwakar	240103049
Kaluram Bhakar	240103059

COURSE: Design of Machine Elements (ME 3104)

DESIGN STANDARDS: IS 2693 / DIN 116 / ISO Standards

FAILURE CRITERION: Maximum Shear Stress Theory (Tresca)

KEY STANDARD: IS 2048 (Sunk Key Proportions)

IMPLEMENTATION: HTML/CSS/JS Web App + Python CLI + C++ CLI

LIVE WEBSITE: aayuuraj.github.io/FlangeCoupling

REPOSITORY: github.com/aayuuraj/FlangeCoupling

November 2026

Table of Contents

1. Introduction & Objective

2. Theory & Design Fundamentals

- 2.1 What is a Rigid Flanged Coupling?
- 2.2 Failure Criterion — Maximum Shear Stress Theory (Tresca)
- 2.3 Design Constraints & Boundary Conditions

3. Design Formulas & Calculation Procedure

- 3.1 Shaft Design
- 3.2 Torsional Rigidity Check
- 3.3 Sunk Key Design (IS 2048)
- 3.4 Hub & Flange Design (IS 2693)
- 3.5 Fastener (Bolt) Array Design

4. Material Database & Properties

5. Design Example — Numerical Calculation

6. Software Implementation

- 6.1 Architecture Overview
- 6.2 Web Application (HTML/CSS/JavaScript)
- 6.3 Python CLI Program
- 6.4 C++ CLI Program

7. CAD Drawing & Visualization

8. Results & Verification

9. Conclusion

10. References

1. Introduction & Objective

A **coupling** is a mechanical device used to connect two rotating shafts for the purpose of transmitting power and torque. Among the various types of couplings, the **Rigid Flanged Coupling (Protected Type)** is one of the most widely used in industrial machinery due to its simplicity, reliability, and capacity to transmit large torques without relative motion between the connected shafts.

Objective

The objectives of this project are:

1. **Design** a complete Rigid Flanged Coupling using the Maximum Shear Stress Theory (Tresca Criterion) in compliance with IS 2693, DIN 116, and IS 2048 standards.
2. **Develop** an interactive CAD workstation (web-based) that takes input torque, material properties, and design constraints, and computes all output coupling dimensions with live stress verification and SVG technical drawings.
3. **Implement** the same calculation engine in Python and C++ for cross-platform portability and evaluation.
4. **Verify** all design outputs with safety factor checks, rigidity checks, and constraint compliance.

2. Theory & Design Fundamentals

2.1 What is a Rigid Flanged Coupling?

A rigid flanged coupling consists of two flanges, each keyed to the respective shaft, and connected together by a set of bolts arranged on a pitch circle. The "protected" type features a raised rim on the outer periphery of each flange that covers the bolt heads and nuts, thereby protecting them from damage and preventing accidental contact.

The main components of a rigid flanged coupling are:

- **Shaft:** The rotating member that transmits the torque. Its diameter is the fundamental design parameter from which all other dimensions are derived.
- **Hub:** The enlarged cylindrical portion of the flange that fits over the shaft. It transmits torque from the shaft to the flange via the key.
- **Flange:** The disc portion that mates with the other half of the coupling. Typically made of cast iron (FG 200 or FG 260).
- **Sunk Key:** A rectangular key fitted into keyways cut in both the shaft and hub. It transmits torque via shear and crushing.
- **Bolts:** A set of bolts arranged uniformly on a pitch circle diameter (PCD) to clamp the two flanges together and transmit torque via shear.
- **Protective Rim:** A raised ring on the outer circumference of the flange that encloses the bolt heads.

2.2 Failure Criterion — Maximum Shear Stress Theory (Tresca)

For ductile materials such as carbon steels and alloy steels used for the shaft, key, and bolts, the **Maximum Shear Stress Theory (MSST / Tresca Criterion)** is the appropriate failure criterion. This theory states that failure (yielding) occurs when the maximum shear stress in the member equals the shear stress at yield in a simple tension test:

TRESCA CRITERION — PERMISSIBLE STRESSES

Allowable Shear Stress:

$$\tau_{allow} = S_{yt} / (2 \times FoS)$$

Allowable Crushing (Compressive) Stress:

$$\sigma_{c,allow} = S_{yt} / FoS$$

Where:

S_{yt} = Tensile yield strength of the material (MPa)

FoS = Factor of Safety (typically 2.5 – 4.0 for static design)

For brittle materials like Grey Cast Iron used for flanges, the permissible shear stress is a stated design limit: $\boldsymbol{\tau_{ci} = 15.0}$ **MPa**.

2.3 Design Constraints & Boundary Conditions

Two design constraints govern the overall feasibility:

CONSTRAINT	SYMBOL	NATURE	DESCRIPTION
Max Shaft Diameter	D_{max}	Spatial Envelope (Input)	Maximum allowable shaft diameter dictated by housing/clearance. If calculated D > D_{max}, a stronger material must be selected.
Max Angle of Twist	θ_{max}	Rigidity Standard (Input)	Maximum torsional twist per meter span (standard: 1.0° per 1000 mm). If $\theta > \theta_{max}$, shaft is upsized for stiffness.

Key Design Principle: The constraints D_{max} and θ_{max} are user-specified boundary limits. The actual shaft diameter D and angle of twist θ are mechanically calculated from the applied Torque T and material properties (S_{yt}, G). If θ exceeds θ_{max}, D is automatically recalibrated upward. If D exceeds D_{max}, the user must select a stronger material.

3. Design Formulas & Calculation Procedure

The design proceeds step-by-step from the shaft diameter as the primary output, and all subsequent component dimensions are proportional functions of D. This section documents every formula used in the calculation engine.

3.1 Shaft Design (Torsional Shear Strength)

The shaft diameter is determined from the torsion equation for a solid circular shaft:

SHAFT DIAMETER FROM TORSIONAL STRENGTH

$$T = (\pi / 16) \times \tau_{allow} \times d^3$$

Solving for d:

$$\mathbf{d = [\ (16 \times T) / (\pi \times \tau_{allow})\ ]^{1/3}}$$

Where:

- T = Transmitted torque (N·mm)
- $\tau_{allow} = S_{yt} / (2 \times FoS)$ — Permissible shear stress (MPa)

The calculated d is rounded up to the nearest standard diameter (multiple of 5 mm, minimum 20 mm):

$$\mathbf{D = max(20,\ ceil(d / 5) \times 5)}$$

The induced torsional shear stress in the adopted shaft is then verified:

INDUCED SHEAR STRESS VERIFICATION

$$\mathbf{\tau_{shaft} = (16 \times T) / (\pi \times D^3) \quad \leq \quad \tau_{allow}}$$

3.2 Torsional Rigidity Check

Even if the shaft is strong enough, it must be stiff enough. The angle of twist over a standard 1-

meter span is checked:

TORSIONAL RIGIDITY — ANGLE OF TWIST

Polar Moment of Inertia: $J = (\pi / 32) \times D^4$

Angle of Twist: $\theta = (T \times L) / (J \times G) \times (180 / \pi)$ degrees

Where:

L = 1000 mm (standard verification span)

G = Shear modulus of the shaft material (MPa)

Criterion: $\theta \leq \theta_{max}$ (standard: 1.0° per meter)

If $\theta > \theta_{max}$ — Rigidity Recalibration:

$D_{twist} = [\ (32 \times T \times L) / (\pi \times G \times \theta_{max,rad})\ ]^{1/4}$

$D = \max(D_{strength}, \text{ceil}(D_{twist} / 5) \times 5)$

3.3 Sunk Key Design (IS 2048)

The rectangular sunk key dimensions are proportioned relative to the shaft diameter per IS 2048:

SUNK KEY PROPORTIONS (IS 2048)

Key Width: $w = \text{ceil}(D / 4)$ mm

Key Height: $h = \text{ceil}(D / 4)$ mm

Key Length: $l = 1.5 \times D$ mm

The key must withstand both shearing and crushing:

KEY STRESS VERIFICATION

Shear failure:

$$T = l \times w \times \tau_k \times (D / 2)$$

$$\tau_k = (2 \times T) / (D \times w \times l) \quad \leq \quad \tau_{allow}$$

Crushing failure:

$$T = l \times (h / 2) \times \sigma_{ck} \times (D / 2)$$

$$\sigma_{ck} = (4 \times T) / (D \times h \times l) \quad \leq \quad \sigma_{c,allow}$$

3.4 Hub & Flange Design (IS 2693 / DIN 116)

All flange and hub dimensions are proportional to the shaft diameter D:

Table 3.1 — Flange & Hub Proportions (IS 2693)

COMPONENT	SYMBOL	FORMULA	DESCRIPTION
Hub Outer Diameter	D_1	$2 \times D$	Outer diameter of the cylindrical hub
Hub Length	L_{hub}	$1.5 \times D$	Axial length of the hub
Pitch Circle Diameter	D_2	$3 \times D$	Bolt arrangement circle diameter
Flange Outside Diameter	D_3	$4 \times D$	Outer diameter of the flange disc
Flange Thickness	t_f	$0.5 \times D$	Axial thickness of the flange disc
Protective Rim Thickness	t_p	$0.25 \times D$	Raised ring that covers bolt heads

Two shear stress checks are performed on the cast iron flange body:

FLANGE BODY STRESS VERIFICATION

Hub Torsional Shear (Hollow Cylinder):

$$\tau_{hub} = (16 \times T \times D_1) / [\ \pi \times (D_1^4 - D^4)\ ] \quad \leq \quad 15.0 \text{ MPa}$$

Flange Circumferential Shear (at hub junction):

$$\tau_f = (2 \times T) / (\pi \times D_1^2 \times t_f) \quad \leq \quad 15.0 \text{ MPa}$$

3.5 Fastener (Bolt) Array Design

BOLT COUNT & SIZING

Number of Bolts:

$$i = \text{ceil}(D / 50 + 3)$$

Bolt Core Diameter from Shear Strength:

$$T = i \times (\pi/4) \times d_b^2 \times \tau_{allow} \times (D_2 / 2)$$

$$d_b = \sqrt{[\ (8 \times T) / (i \times \pi \times D_2 \times \tau_{allow})\ ]}$$

The calculated core diameter is rounded up and matched to the nearest standard ISO metric bolt size (M6, M8, M10, M12, M14, M16, M18, M20, M22, M24, M27, M30, M36, M42).

Empirical Proportion Check:

$$d_b \in [\ 0.40 \times D / \sqrt{i}\ \ ,\ \ 0.45 \times D / \sqrt{i}\ ]$$

Bolt Shear Stress Verification:

$$\tau_{bolt} = (8 \times T) / (i \times \pi \times D_2 \times d_b^2) \quad \leq \quad \tau_{allow}$$

4. Material Database & Properties

4.1 Ductile Steel Grades (Shaft, Key & Bolts)

Table 4.1 — Ductile Steel Material Database

#	MATERIAL GRADE	CATEGORY	S_{YT} (MPA)	S_{UT} (MPA)	G (MPA)
1	20C8	Low Carbon Steel	260	440	79,300
2	30C8	Mild Carbon Steel	310 / 400*	500	79,000
3	40C8	Plain Carbon Steel	380	580	79,000
4	45C8	Medium Carbon Steel	360 / 380*	600 / 630*	79,000
5	50C4	Carbon Steel	460	660	79,000
6	55C8	Plain Carbon Steel	460	720	79,000
7	40Cr4	Alloy Steel	490	690	80,000
8	40Cr4Mo2	Alloy Steel	490	700	80,000
9	35Ni5Cr2	Alloy Steel	490	690	80,000

* Minor variation between Python CLI and Web App databases.

4.2 Brittle Materials — Cast Iron (Flange Body)

Table 4.2 — Grey Cast Iron Material Database

#	MATERIAL GRADE	S_{UT} (MPA)	G (MPA)	$T_{CI,ALLOW}$ (MPA)
1	FG 200	200	41,000	15.0 (Stated Limit)
2	FG 260	260	41,000	15.0 (Stated Limit)

5. Design Example — Numerical Calculation

A complete worked example using default inputs:

5.1 Given Data

PARAMETER	VALUE
Transmitted Torque (T)	6,207 N·m = 6,207,000 N·mm
Factor of Safety (FoS)	3.0
Shaft Material	40C8 Plain Carbon Steel (S_{yt} = 380 MPa, G = 79,000 MPa)
Flange Material	FG 200 Grey Cast Iron (S_{ut} = 200 MPa)
Max Shaft Diameter (D_{max})	150 mm
Max Angle of Twist (θ_{max})	1.0° per 1000 mm span

5.2 Step-by-Step Calculations

Step 1: Permissible Stresses

> τ_{allow} = 380 / (2 × 3.0) = **63.33 MPa**
> $\sigma_{c,allow}$ = 380 / 3.0 = **126.67 MPa**

Step 2: Shaft Diameter (Strength)

> d = $[(16 \times 6{,}207{,}000) / (\pi \times 63.33)]^{1/3}$ = $[99{,}312{,}000 / 198.97]^{1/3}$ = $[499{,}117.3]^{1/3}$
> d = **79.3 mm** → Standardized D = **80 mm**

Step 3: Torsional Rigidity Check

$J = (\pi / 32) \times 80^4 = (\pi / 32) \times 40{,}960{,}000 = \mathbf{4{,}021{,}239\ mm^4}$

$\theta = (6{,}207{,}000 \times 1000) / (4{,}021{,}239 \times 79{,}000) \times (180 / \pi)$

$\theta = 6{,}207{,}000{,}000 / 317{,}677{,}881{,}000 \times 57.296$

$\theta = 0.01954 \times 57.296 = \mathbf{1.120°}$

Since θ (1.120°) > θ_{max} (1.0°) → **Shaft must be upsized for rigidity**

$D_{twist} = [(32 \times 6{,}207{,}000 \times 1000) / (\pi \times 79{,}000 \times 0.01745)]^{0.25}$

$D_{twist} = [198{,}624{,}000{,}000 / 4{,}328.3]^{0.25} = [45{,}889{,}370.5]^{0.25}$

$D_{twist} = \mathbf{82.3\ mm}$ → Standardized D = **85 mm**

Step 4: Key Dimensions (IS 2048)

$w = \text{ceil}(85 / 4) = \mathbf{22\ mm}$

$h = \text{ceil}(85 / 4) = \mathbf{22\ mm}$

$l = 1.5 \times 85 = \mathbf{127.5\ mm}$

$\tau_k = (2 \times 6{,}207{,}000) / (85 \times 22 \times 127.5) = 12{,}414{,}000 / 238{,}425$

$= \mathbf{52.1\ MPa} \leq 63.33$ ✓

$\sigma_{ck} = (4 \times 6{,}207{,}000) / (85 \times 22 \times 127.5) = 24{,}828{,}000 / 238{,}425$

$= \mathbf{104.1\ MPa} \leq 126.67$ ✓

Step 5: Flange & Hub Dimensions

$D_1 = 2 \times 85 = \mathbf{170\ mm}$ | $L_{hub} = 1.5 \times 85 = \mathbf{127.5\ mm}$

$D_2 = 3 \times 85 = \mathbf{255\ mm}$ | $D_3 = 4 \times 85 = \mathbf{340\ mm}$

$t_f = 0.5 \times 85 = \mathbf{42.5\ mm}$ | $t_p = 0.25 \times 85 = \mathbf{21.25\ mm}$

$$\tau_{hub} = (16 \times 6{,}207{,}000 \times 170) / [\pi \times (170^4 - 85^4)]$$
$$= 16{,}883{,}040{,}000 / [\pi \times (835{,}210{,}000 - 52{,}200{,}625)]$$
$$= 16{,}883{,}040{,}000 / 2{,}458{,}690{,}547$$
$$= \mathbf{6.87\ MPa} \leq 15.0 \checkmark$$

$$\tau_f = (2 \times 6{,}207{,}000) / (\pi \times 170^2 \times 42.5) = 12{,}414{,}000 / 3{,}855{,}761$$
$$= \mathbf{3.22\ MPa} \leq 15.0 \checkmark$$

Step 6: Bolt Array

$i = \text{ceil}(85 / 50 + 3) = \text{ceil}(4.7) = \mathbf{5\ Bolts}$

$$d_b = \sqrt{[(8 \times 6{,}207{,}000) / (5 \times \pi \times 255 \times 63.33)]}$$
$$= \sqrt{[49{,}656{,}000 / 253{,}711.3]} = \sqrt{[195.72]}$$
$$= \mathbf{13.99\ mm} \rightarrow \text{Rounded to 14 mm} \rightarrow \text{Adopted: } \mathbf{M16}$$

Empirical check: $[0.40 \times 85 / \sqrt{5},\ 0.45 \times 85 / \sqrt{5}] = [15.2\ mm, 17.1\ mm] \checkmark$

$$\tau_{bolt} = (8 \times 6{,}207{,}000) / (5 \times \pi \times 255 \times 14^2) = 49{,}656{,}000 / 786{,}072$$
$$= \mathbf{63.2\ MPa} \leq 63.33 \checkmark$$

5.3 Final Results Summary

Table 5.1 — Complete Output Dimensions

COMPONENT	PARAMETER	VALUE	STATUS
Shaft	D	85.0 mm	PASS ($D \leq D_{max}$)
Shaft	θ (twist)	0.878°	PASS ($\theta \leq 1.0°$)
Key	w × h × l	22 × 22 × 127.5 mm	PASS
Hub	D_1, L_{hub}	170, 127.5 mm	PASS
Flange	D_2, D_3, t_f, t_p	255, 340, 42.5, 21.25 mm	PASS
Bolts	5 × M16	Core 14 mm on PCD 255 mm	PASS

6. Software Implementation

6.1 Architecture Overview

The project delivers three independent implementations of the same calculation engine, each targeting a different execution environment:

Table 6.1 — Implementation Matrix

PLATFORM	FILE	LANGUAGE	PURPOSE
Web Browser	`index.html`	`HTML + CSS + JavaScript`	Interactive CAD workstation with live SVG drawings, sliders, themes, and export
Terminal / CLI	`coupling_design.py`	`Python 3`	Interactive CLI with prompts, ANSI-formatted reports, ASCII CAD schematic, file export
Terminal / CLI	`coupling_design.cpp`	`C++17`	Compiled CLI with calculation engine, ASCII schematic, and report file export

6.2 Web Application (HTML/CSS/JavaScript)

The web application is a **single standalone HTML file** (~2400 lines) requiring no build tools, frameworks, or external dependencies (except Google Fonts). It features:

- **Three-Column Workspace Layout:** Input panel (left), CAD viewport (center), Output dimensions panel (right).
- **Live Reactive Calculations:** All sliders, dropdowns, and number inputs trigger instant recalculation and re-rendering.
- **SVG CAD Drawing Engine:** Two views — Full Section A-A (longitudinal cross-section) and Flange Elevation (end view) — with ISO hatching patterns, centerlines, dimension arrows, and bolt arrays. A dual drafting sheet mode shows both views.
- **Three Themes:** Studio Light (SolidWorks / Fusion 360 style), Classic Blueprint

(AutoCAD blue), and Titanium Dark.

- **Torque Calculator:** Primary direct torque input with secondary power/speed/overload derivation: $T = (P \times 60{,}000) / (2\pi \times N) \times K_s$.
- **Constraint Telemetry Cards:** Live comparison grids showing specified limits vs. mechanically calculated values with PASS/FAIL/UPSIZED badges.
- **Calculation Dossier Modal:** Formal engineering report popup with Copy Plain Text and Print/Export PDF buttons.
- **Mouse-Wheel Zoom:** Canvas zoom 60%–250% with "Zoom to Fit (100%)" reset button.

Key JavaScript Function: `runCalculations()`

This is the core calculation engine (equivalent logic across all three implementations):

```
function runCalculations() {
  // 1. Read inputs: T_Nm, FoS, Material selection
  // 2. Compute permissible stresses: τ_allow, σ_c_allow
  // 3. Shaft diameter from strength: d = [(16T)/(πτ)]^(1/3)
  // 4. Rigidity check: θ = (T·L)/(J·G)·(180/π)
  //    → If θ > θ_max: upsize D via D_twist formula
  // 5. Key dimensions: w, h, l with τ_k, σ_ck checks
  // 6. Hub/Flange dimensions: D1, D2, D3, tf, tp
  //    with τ_hub, τ_f checks against 15 MPa
  // 7. Bolt array: count i, core dia, adopted metric size
  //    with τ_bolt check
  // 8. Store results → updateOutputUI() → renderCadSchematic()
}
```

6.3 Python CLI Program

File: `coupling_design.py` (346 lines). Uses the `FlangedCouplingDesign` class to encapsulate all calculations. Features an interactive prompt loop via `safe_input()`, material database selection menus, and automatic export to `coupling_report.txt`.

Running the Python Program

```
$ python3 coupling_design.py

# Interactive prompts for:
# - Torque input method (direct / power+speed)
# - Factor of Safety
# - Shaft material (8 ductile steels)
# - Flange material (2 cast irons)
# - D_max and θ_max constraints
```

6.4 C++ CLI Program

File: `coupling_design.cpp` (234 lines). A single-file C++17 program implementing the identical calculation workflow using standard library headers (`<cmath>`, `<vector>`, `<fstream>`, `<iomanip>`). Exports a summary to `coupling_report_cpp.txt`.

Compiling and Running

```
$ g++ -O2 -std=c++17 coupling_design.cpp -o coupling_design
$ ./coupling_design
```

7. CAD Drawing & Visualization

7.1 Section View (Longitudinal Cross-Section)

The section view shows the coupling cut along the axial plane (Section A-A), displaying:

- Left and right flanges with ISO 45° hatching (opposite directions per convention)
- Shaft through the center with steel hatching
- Sunk keys on top of each shaft half
- Bolt cross-sections at the pitch circle diameter
- Centerline (red, long-dash-short-dash pattern)
- PCD reference lines (amber, dashed)
- Tiered dimension lines with arrows and labels for D, D1, D2, D3, tf, tp, Hub L, Key l

7.2 ASCII Cross-Section Schematic

Both the Python and C++ programs generate an ASCII-art technical schematic in the terminal:

```
                    +-----------------------------------------+
                    |       OUTER FLANGE RIM  (D3 = 340.0 mm)      |
                    +----------+----------------------+----------+
                    |          |  BOLT PCD (D2 = 255.0)|          |
                    |          |      [ (O) 5x M16 ]   |          |
                    |          +-----------------------+          |
                    |   HUB    |   SUNKEY  w=22 h=22   |   HUB    |
====================| (D1=170) |   +---------------+   | (D1=170) |==================
 SHAFT (D =  85.0)  |          |   |===============|   |          |  SHAFT (D =  85.
====================|          |   +---------------+   |          |==================
                    |          +-----------------------+          |
                    |          |      [ (O) BOLT ]     |          |
                    |          |  FLANGE THICKNESS     |          |
                    +----------+-- (tf =  42.5 mm) --+----------+
                    |       OUTER FLANGE RIM  (D3 = 340.0 mm)      |
                    +-----------------------------------------+
                    <--Lhub=127.5->|<-tf=42.5->|

  * NOTE: ALL DIMENSIONS ARE IN mm
```

7.3 Flange Elevation View (End View)

The end view renders a front-facing view of the flange showing:

- Concentric circles for D_3 (outer flange), protective rim, PCD (dashed), D_1 (hub), and D (shaft bore)
- Bolt positions uniformly distributed on the PCD
- Key slot at top of shaft bore
- Horizontal and vertical centerlines

8. Results & Verification

8.1 Stress Verification Summary

Table 8.1 — Complete Stress Verification (Default Example)

COMPONENT	STRESS TYPE	INDUCED	PERMISSIBLE	FOS (ACTUAL)	STATUS
Shaft	Torsional Shear (τ)	51.5 MPa	63.33 MPa	3.69	PASS ✓
Key	Shear (τ_k)	52.1 MPa	63.33 MPa	3.65	PASS ✓
Key	Crushing (σ_{ck})	104.1 MPa	126.67 MPa	3.65	PASS ✓
Hub	Torsional Shear (τ_{hub})	6.87 MPa	15.0 MPa	—	PASS ✓
Flange	Shear at Hub Junction (τ_f)	3.22 MPa	15.0 MPa	—	PASS ✓
Bolts	Direct Shear (τ_{bolt})	63.2 MPa	63.33 MPa	3.01	PASS ✓

8.2 Constraint Compliance

Table 8.2 — Constraint Verification

CONSTRAINT	SPECIFIED LIMIT	CALCULATED VALUE	MARGIN	STATUS
Spatial Envelope (D_{max})	150.0 mm	85.0 mm (D)	+65.0 mm	PASS ✓
Torsional Rigidity (θ_{max})	1.000° / m	0.878° / m	+0.122°	PASS (Upsized from 80→85 mm)

8.3 Cross-Platform Verification

All three implementations (Web/JS, Python, C++) produce identical output dimensions when given the same inputs, confirming the calculation engine's correctness and cross-platform portability.

9. Conclusion

This project successfully designed a Rigid Flanged Coupling (Protected Type) using the Maximum Shear Stress Theory (Tresca Criterion) in compliance with IS 2693, DIN 116, and IS 2048 standards. The complete design workflow — from applied torque to all coupling dimensions with stress verification — was implemented as an interactive web-based CAD workstation and replicated in Python and C++ for portability.

Key achievements:

1. **Complete Static Design:** All four component groups (Shaft, Key, Flange/Hub, Bolts) are designed and verified with their respective stress checks.
2. **Dual Constraint System:** The spatial envelope (D_{max}) and torsional rigidity (θ_{max}) constraints are properly integrated with automatic shaft recalibration.
3. **Interactive Visualization:** Live SVG technical drawings with ISO hatching, dimension annotations, and multiple CAD views provide immediate visual feedback.
4. **Materials Database:** Nine ductile steel grades and two grey cast iron grades with full mechanical properties enable comparative material studies.
5. **Production-Ready Deployment:** The web application is deployed live at **aayuuraj.github.io/FlangeCoupling** and works entirely client-side with zero server dependencies.

10. References

1. IS 2693:1980 — *Specification for Protected Type Rigid Flanged Couplings*, Bureau of Indian

Standards.

2. DIN 116 — *Rigid Flanged Shaft Couplings*, Deutsches Institut für Normung.
3. IS 2048:1983 — *Parallel Keys and Keyways — Dimensions and Tolerances*, Bureau of Indian Standards.
4. IS 1570:1978 — *Schedules for Wrought Steels*, Bureau of Indian Standards.
5. V.B. Bhandari, *Design of Machine Elements*, 4th Edition, McGraw-Hill Education.
6. Joseph E. Shigley, *Mechanical Engineering Design*, 10th Edition, McGraw-Hill.
7. R.S. Khurmi and J.K. Gupta, *A Textbook of Machine Design*, S. Chand Publishing.

— End of Project Report —

Generated for: Rigid Flanged Coupling CAD & Design Workstation

Live: aayuuraj.github.io/FlangeCoupling | Source: github.com/aayuuraj/FlangeCoupling